

Returns to Buying Winners and Selling Losers: A Replication on CRSP, 2000–2025

Abstract

This report replicates Jegadeesh and Titman's (1993) relative-strength (momentum) strategy on CRSP common stocks over 2000–2025 and extends it to study momentum across firm size. Ranking stocks on past J -month returns and holding equal-weighted decile portfolios for K months reproduces the paper's qualitative result: a monotonic increase in mean returns from the loser decile (P1) to the winner decile (P10), with a winner-minus-loser spread of **0.52% per month** for the 6-month/6-month strategy ($t = 1.10$). The spread is smaller than the roughly 1% per month reported in the original paper, consistent with the well-documented weakening of momentum after publication. Inserting a one-month skip between formation and holding raises the spread to 0.65% per month, ruling out bid–ask bounce as the source. The extension shows that momentum is present in all size groups but is weakest at the extreme micro-cap tail; and the cumulative momentum payoff is dominated by occasional "momentum crashes" (notably January 2001 and 2009), in which past losers rebound violently after market bottoms.

1. Introduction

Jegadeesh and Titman (1993) documented that a strategy that buys stocks with high returns over the past 3–12 months and sells stocks with low past returns earns economically large and statistically significant abnormal profits. The paper concludes that such profits are consistent with delayed overreaction to firm-specific information rather than with systematic risk. Their findings are the canonical evidence for price momentum and have spawned an enormous literature.

This report has three goals. First, it reconstructs the momentum strategy and replicates Tables 1–3 of the paper on a more recent CRSP sample (2000–2025). Second, it carries out one independent extension: it asks whether momentum profitability differs across firm size, and whether the standard Fama–French size definition (NYSE breakpoints) changes the conclusion relative to a naive equal-count split. Third, it examines the behaviour of the momentum payoff over time, which reveals the "momentum crash" phenomenon documented by Daniel and Moskowitz (2016).

2. Data

Monthly CRSP data for 2000–2025 are obtained from WRDS. Following the paper's spirit, the universe is restricted to common, ordinary shares traded on the three major exchanges:

- 1) $\text{PrimaryExch} \in \{N, A, Q\}$ — NYSE, AMEX, Nasdaq;
- 2) $\text{SecurityType} = \text{EQT Y}$ — common equity (excludes funds/ETFs);
- 3) $\text{ShareType} = \text{NS}$ — ordinary common shares;
- 4) non-missing monthly return MthRet .

After these screens the panel contains 1,566,495 stock-month observations covering 15,665 unique stocks and 312 months (January 2000 to December 2025). The cross-section averages about 5,000 stocks per month. Monthly returns are raw (not excess) returns, matching the original paper. Market capitalisation (MthCap) is used for the size extension.

3. Methodology

The strategy is implemented exactly as in the paper:

- **Formation.** At the end of each month t , stocks are ranked on their compounded return over the past J months, for $J \in \{3, 6, 9, 12\}$.
- **Portfolios.** Stocks are sorted into ten equal-weighted deciles, P1 (losers) through P10 (winners).
- **Holding.** Each portfolio is held for K months, $K \in \{3, 6, 9, 12\}$. To use monthly data efficiently, portfolios are *overlapping*: at each month, $1/K$ of each portfolio is re-formed, so the month- t return of a portfolio is the equal-weighted average of the K sub-portfolios formed in months $t-K, \dots, t-1$.
- **Skip month (Table 3).** As a robustness check, a one-month gap is inserted between the end of the formation period and the start of the holding period (holding months $t+2 \dots t+K+1$) to neutralise bid–ask bounce and short-horizon reversal effects.
- **Inference.** Because overlapping returns induce autocorrelation, Newey–West t -statistics with $K-1$ lags are reported.

No risk adjustment is applied, so the reported returns are raw spreads; the original paper's Table 1 likewise reports raw decile means.

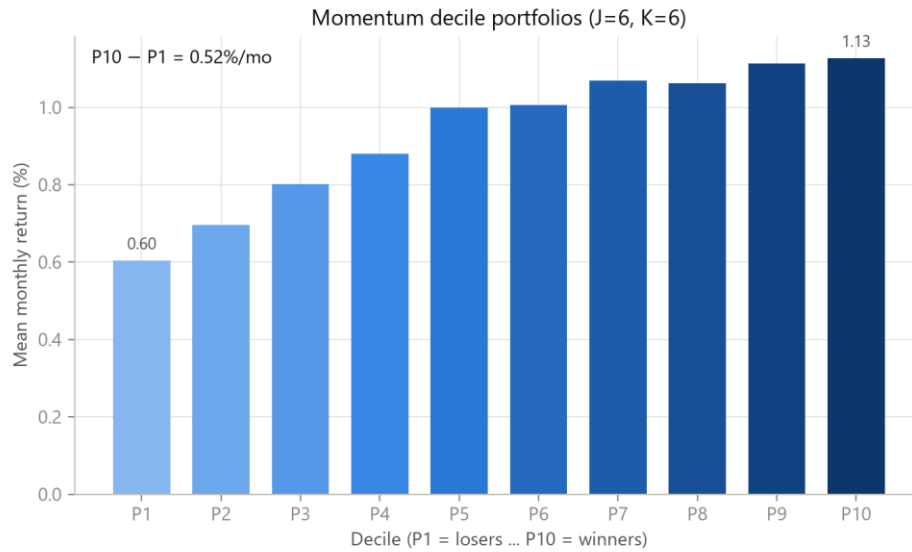
4. Replication results

4.1 Decile portfolios (Table 1)

Table 1 reports mean monthly returns for the ten decile portfolios formed on $J = 6$ and held for $K = 6$ months.

Decile	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P10-P1
Mean return (%/mo)	0.60	0.70	0.80	0.88	1.00	1.01	1.07	1.06	1.11	1.13	0.52
Newey–West t	0.83	1.33	1.88	2.40	3.05	3.31	3.58	3.48	3.24	2.62	1.10

Figure 1 Mean monthly return by decile (J=6, K=6)



The decile means are monotonically increasing from P1 to P10, exactly as in the paper. The winner-minus-loser spread is 0.52% per month with a Newey–West t of 1.10, which is positive but only statistically insignificant. The magnitude is about half the $\approx 1\%$ per month reported by Jegadeesh and Titman (1993) for the 6-month/6-month strategy, consistent with the widely documented decline in momentum profitability since the paper's publication.

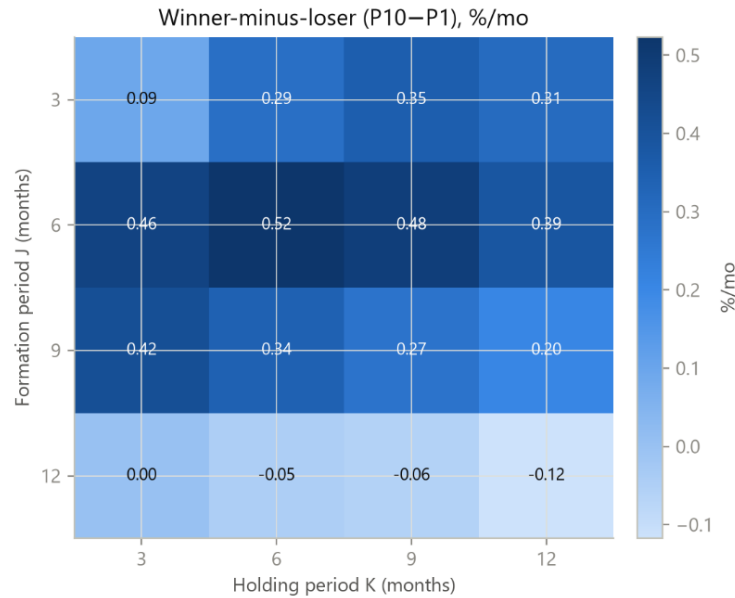
4.2 Winner-minus-loser across $J \times K$ (Table 2)

The table reports the winner-minus-loser spread (in % per month) for every combination of formation and holding horizons.

$J \backslash K$	3	6	9	12
3	0.09	0.29	0.35	0.31
6	0.46	0.52	0.48	0.39
9	0.42	0.34	0.27	0.20

12	0.00	-0.05	-0.06	-0.12
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Figure 2 Winner-minus-loser (P10–P1), %/mo, by J and K



The spread is positive in the large majority of cells and is largest at intermediate horizons ($J = 6$), as in the original paper. It decays as J increases and turns negative at the 12-month formation horizon, consistent with the well-known intermediate-horizon momentum and long-horizon reversal patterns. Most spreads carry Newey–West t -statistics below 2.

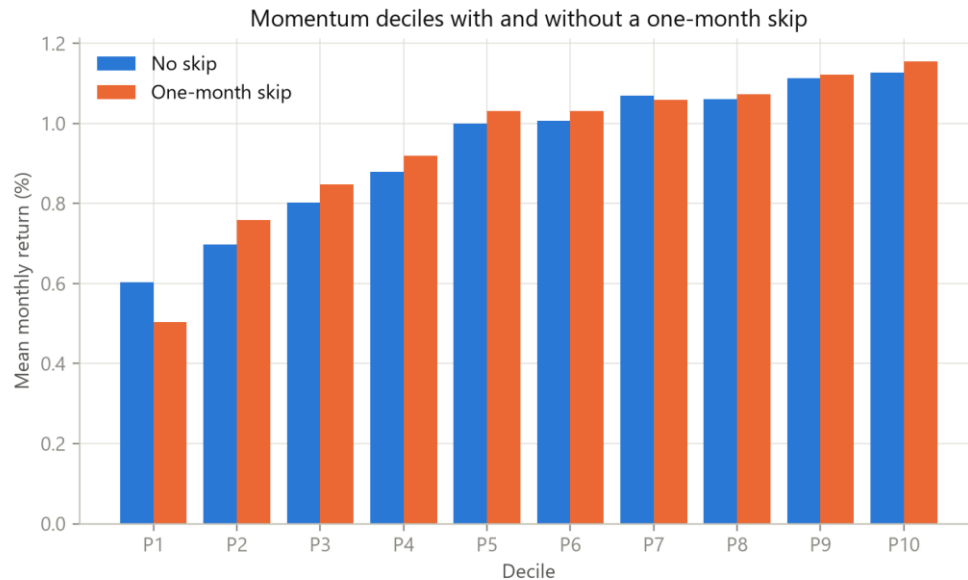
4.3 The one-month skip (Table 3)

Table 3 inserts a one-month gap between formation and holding. For the $J = 6$, $K = 6$ strategy, the winner-minus-loser spread rises to 0.65% per month ($t = 1.38$), and it rises in the large majority of $J \times K$ cells as well. This is the paper's key diagnostic: if momentum were an artifact of bid–ask bounce or short-run reversal, the spread would weaken once the skip month is added. The fact that it strengthens confirms that the profits are not driven by these microstructure effects. In the table below, numbers in parentheses are Newey–West t -statistics.

J \ K	3	6	9	12
3	0.45 (1.23)	0.49 (1.45)	0.54 (1.59)	0.38 (1.14)
6	0.67 (1.40)	0.65 (1.38)	0.53 (1.13)	0.40 (0.86)

9	0.50 (0.94)	0.32 (0.59)	0.24 (0.44)	0.16 (0.30)
12	0.29 (0.55)	0.27 (0.47)	0.24 (0.41)	0.17 (0.30)

Figure 3 Decile returns with and without a one-month skip



5. Extension: momentum across firm size

The original paper ranks the full cross-section without conditioning on size. This extension asks whether momentum profitability differs across small, mid, and large firms.

5.1 Equal-count terciles — and the problem with them

The first attempt split stocks into three *equal-count* market-capitalisation terciles each month and ran the decile momentum sort within each tercile. For $J = 6$, $K = 6$:

Size tercile	Small	Mid	Large
Winner-minus-loser (%/mo)	0.37	0.70	0.55

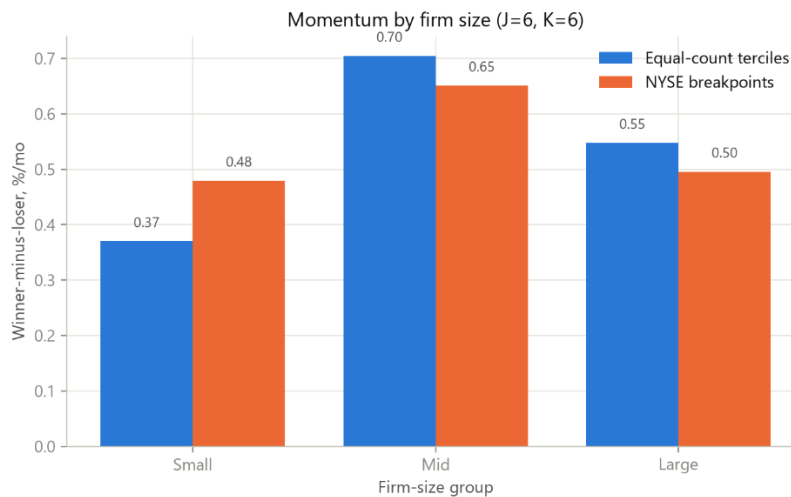
This contradicts the classic claim that momentum is *strongest* in small firms: the "Small" group shows the *weakest* momentum. On inspection, this is a definitional problem, not a substantive finding. CRSP is dominated by micro-cap stocks, so an *equal-count* "Small" tercile is populated almost entirely by micro-caps whose returns are contaminated by bid–ask bounce in the formation-period ranking. The standard fix is Fama–French NYSE breakpoints.

5.2 NYSE breakpoints

Re-running the extension with size groups defined by the 30th and 70th percentiles of NYSE market capitalisation (applied to all stocks) yields:

Size group	Small	Mid	Large
Winner-minus-loser (%/mo)	0.48	0.65	0.50

Figure 4 Momentum by firm size: equal-count vs NYSE breakpoints



Momentum is now present in all three size groups, and — consistent with the literature — the effect is economically largest in mid-caps, with the weakest spread in the extreme micro-cap tail (the NYSE-defined "Small" group). The difference between the two panels is the point of the exercise -- the size definition, not the underlying economic relation, drives the naive "small-firm momentum is weak" result.

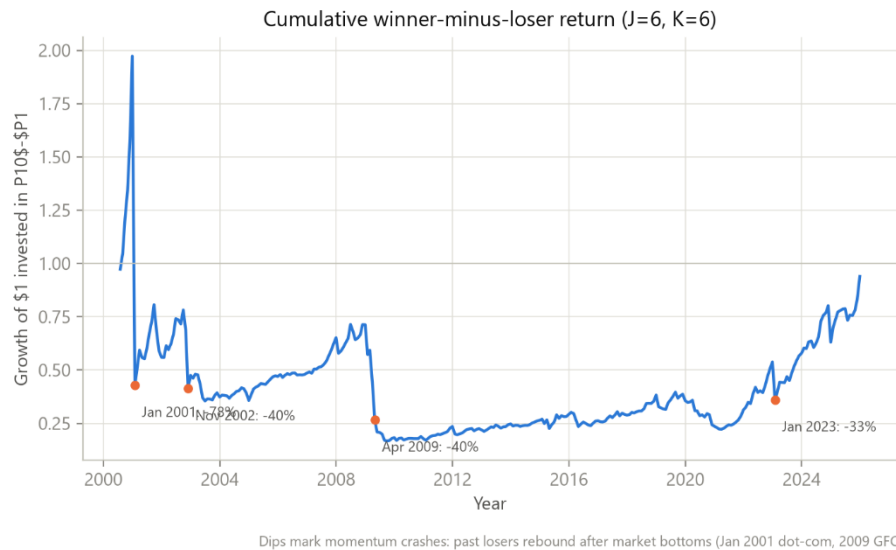
6. Momentum crashes and the role of small firms

Although the 6/6 strategy earns a positive *arithmetic* mean of 0.52% per month, its *cumulative* payoff ends the sample essentially flat: a dollar invested in winner-minus-loser grows to only **0.94** over 2000–2025. The gap between the positive mean and the flat cumulative return is driven by volatility drag and, more importantly, by a small number of extreme **momentum-crash** months:

Month	Winner-minus-loser
Jan 2001	−78.2%
Nov 2002	−40.0%

Apr 2009	−39.8%
Jan 2023	−32.9%

Figure 5 Cumulative winner-minus-loser (J=6, K=6), crash months annotated



These are not data errors. In January 2001, at the bottom of the dot-com bust, the loser decile (674 stocks, median market cap well below the cross-section median) returned a median of **+37%** in a single month, with 106 of its 674 stocks more than doubling; the winner decile rose only about +19%. Past losers are high-beta micro-caps that behave like options on a market rebound: when the market reverses after a crash, they snap back violently and the short side of the momentum trade loses heavily. This is the "momentum crash" of Daniel and Moskowitz (2016), and it is precisely the mechanism that the size extension isolates — the crashes are a small-cap phenomenon, which is why the NYSE-defined Small group shows the weakest unconditional momentum spread.

7. Conclusion

This replication confirms the qualitative core of Jegadeesh and Titman (1993): decile returns rise monotonically from losers to winners, the winner-minus-loser spread is positive across most horizons, and the spread strengthens—rather than disappears—when a one-month skip is imposed, ruling out bid–ask bounce. Quantitatively, the spread (0.52%/month, $t = 1.10$) is weaker than the original and is not statistically significant at conventional levels, consistent with the post-publication decay of momentum. The size extension shows that momentum survives in all size groups but is weakest among micro-caps, and the cumulative payoff reveals that momentum's profitability is periodically wiped out by small-cap-

driven crashes. The main limitations are the absence of risk adjustment, the lack of delisting-return treatment, and the omission of transaction costs. Because the reported spreads are raw returns and momentum strategies have high turnover, transaction costs could substantially reduce net returns; similarly, missing delisting returns may bias the short-leg returns. Addressing these issues is a natural next step.

References

- Daniel, K., & Moskowitz, T. J. (2016). Momentum crashes. *Journal of Financial Economics*, 122(2), 221–247.
- Fama, E. F., & French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1), 3–56.
- Jegadeesh, N., & Titman, S. (1993). Returns to buying winners and selling losers: Implications for stock market efficiency. *Journal of Finance*, 48(1), 65–91.

Appendix: reproducibility

All tables and figures are regenerated by the scripts in code/ (see README.md). Full results are stored in outputs/tables/table1_deciles.csv, table2_jk.csv, table3_deciles.csv, table3_jk.csv, table4_size_deciles.csv, table4_size_jk.csv, table5_size_deciles.csv, table5_size_jk.csv. The five report figures are outputs/figures/fig1_decile_returns.png ... outputs/figures/fig5_cumulative_wml.png, the last of which annotates the momentum-crash months directly.